DRUMPANEL · DESIGN DECISION RECORD

DrumPanel design for construction at detail level D3

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DOCUMENT CONTROL

Document ID	DMP-DDR-006	Version	0.1
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/decisions/0006-design-for-construction-at-d3.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	Details found while modelling every bolt, pin, weld and bearing of option A (kit 1.9.1); made under STANDARDS section 18; no working principle, pitch, scope or safety case changed

0006: Design for construction at detail level D3

- **Date:** 2026-10-24
- **Status:** made under STANDARDS section 18 (Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible") while carrying out the kit 1.9.1 rollout (Amish, 2026-10-24: "I want detailed TRL3 CAD"; "Kit 1.9.1 rollout - proceed to rollout across the other repos"). Amish can reverse any item. None changes what the bench set does, its pitch, its scope or its safety case; none changes a calculation in DMP-CAL-001 or the cost.

Context

Kit 1.9 asks for the model of every option at detail level D3: real stock sections, every bolt, nut, washer, pin and screw with its hole, welds as beads, bearings and bushes, checked for interference (STANDARDS section 24). The station model of 2026-10-03 drew each component as one solid with its main holes. Rebuilding option A part by part in `cad/src/option_a.py` (818 parts, `cad/fidelity.json`) found a few joints that could not be assembled or fastened as drawn or written in the build plan, and several fixings the build plan named but the model did not draw.

Decision

Table 1. Changes made to the model, the BOM text and the build plan.

#	Problem found	Change made	Why this way
Q1	The pivot pin was to take an R-clip below the head plate, but that point is inside the welded drop bar, where no clip can be fitted	The pin is lengthened from 55 to 66 mm and, with the drop bar, cross-drilled 9 mm 15 mm above its lower end, 17 mm below the head plate; an M8 bolt through both holds it; the pin turns in the swivel plate and its collar rides on top	The head hangs from the pin, so the pin must be held below the head plate; a cross bolt is reached from outside the drop bar
Q2	Two M12 foot bolts with washers do not fit on an 80 x 80 mm foot plate round a 50 mm leg	Side frame foot plates 120 x 80 mm, extending inward; both M12 bolts inboard of the leg, their nuts inside the long rail, reached through 24 mm access holes in its bottom wall	Keeps the bolted side frames (DMP-DDR-002, P11) on the same base rails; the rails stand on the floor, so the nuts go inside
Q3	The build plan put the tool shelf bolt holes 550 and 700 mm up from the frame top, above the shelf's back plate	Holes 550 and 700 mm up from the floor, inside the back plate (450 to 760 mm); build plan text corrected	The model's shelf position, which the drawing shows, is kept
Q4	The disc guard's 26 mm shaft hole would not pass the 40 mm disc clamp collar and spacer	42 mm hole through the front and back faces of the guard hood	The clamp and spacer hold the upper disc on its shaft
Q5	Table legs stopped at the lower edge of a box-section frame; real 40 x 40 x 3 mm angles have no lower face there	Legs run up to the underside of the angle frame's top flange and are welded there; each stands on a 50 x 50 x 3 mm cap plate with a welded M10 nut and an M10 levelling screw, as the build plan already said	Gives the weld a face and the levelling screws a nut
Q6	The nip guard lid had no fixing	Four 20 x 20 x 2 mm angle clips, screwed to the side frames with M5 screws and to the lid with M5 bolts	Guards need a tool to remove (PR-010)
Q7	The slide block keeper plates named in the build plan were not drawn; a round captive collar larger than the screw would not pass the 20 mm side plate	Keeper plates of 3 mm sheet with M5 screws on the inside faces; the captive collars on the four screws are flatted to 18 mm	The collars engage the blocks inside the plate thickness
Q8	The cradle roller end discs sat flush with the tube ends, leaving no corner for a fillet weld	Discs set 4 mm into the tube and welded all round	A fillet weld needs a corner
Q9	The shear head bearings were not drawn	Upper housing bored 60 mm for the eccentric bush (3 mm offset) carrying two 6005-2RS bearings; lower housing bored 47 mm for one; drive case plates bored 42 mm for four 6004-2RS bearings; the case closed by three pillar bolts with spacer tubes, its inner plate held by four M8 screws into the upper housing	As in BOM line 10; the case must come apart to fit the chains
Q10	The slip roll crank housing and guards had no fixings	Crank housing a 50 x 6 mm tube with two 25 x 38 x 20 mm bronze bushes; the chain guard held by two M6	Guards need a tool to remove (PR-010)

#	Problem found	Change made	Why this way
		screws into the housing end; the gear guard by a tab and two M6 screws into the side frame	

Consequences

- Design dimensions changed: Q1 (pivot pin 66 mm long, was 55 mm), Q2 (foot plates 120 x 80 mm, were 80 x 80 mm), Q4 (guard hole 42 mm, was 26 mm), Q5 (table legs about 37 mm longer, on levelling screws), Q7 (collars flatted). Masses change by well under 1 kg a station; the calculations in DMP-CAL-001 read the station model and are unchanged.
- Cost: unchanged by these items (USD 3,489 before the second drain stand of STEM fix F2, DMP-DEC-001, which brings it to USD 3,566); the extra fasteners and sheet fall inside BOM line 23 (fasteners, paint and welding consumables, USD 120).
- BOM text updated for lines 9, 12, 13, 15, 18 and 19; build plan DMP-BLD-001 v0.4 updated for Q1, Q2, Q3 and Q6; drawings and build plan pictures regenerated from the D3 model.